Computer Architecture Scribing
Lecture 11

Susi Krishna - cs23btech11062
Lokesh Reddy - cs23btech11011

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1 Combinational Logic:

Combinational Logic always gives the same output for a given input every single time.

1.1 Quick Recap of Previous Lecture

1.1.1 Decoders:

- It is a logic block that has an n-bit input and 2^n outputs, where only one output is true for each input combination.
- Ex: a 2-to-4 decoder has 2 input lines and 4 output lines. Depending on the binary value of the inputs, one of the output lines will be activated (set to 1), while all other lines will be deactivated (set to 0).

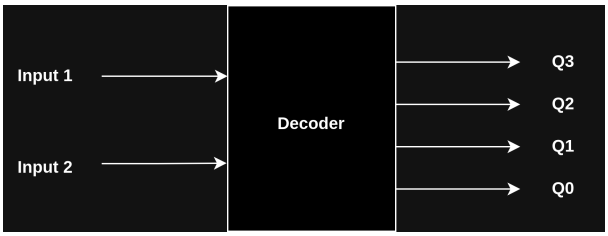


Figure 1: A 2-bit decoder has 4 outputs

- Ex: Glow LED as per switch values

1.1.2 Multiplexors (MUX):

- A multiplexer or MUX is an digital block that selects one of several input and outputs the selected input. It operates just like a multiple-input, single-output switch.
- The output is determined by a control value.
- A 2^n to 1 multiplexer has 2^n input lines and n select lines.

1.1.3 Arithmetic Logic Unit :

- This device performs the arithmetic operations like addition and subtraction as well as logical operations like AND and OR.

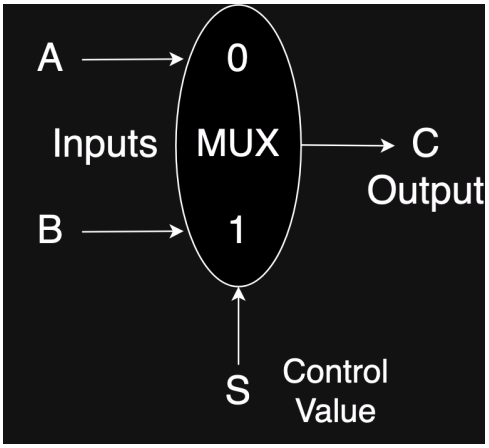


Figure 2: A example of 2^1 to 1 MUX ($C = A.S + B.\overline{S}$)

- Whenever a program requires a mathematical calculation or a logic comparison, the ALU performs the operation.
- Apart from the inputs that need to be operated, an ALU has another input named operation to select the arithmetic or logical operation that needs to be performed.

2 Sequential Logic:

There are few operations that combinational logic can't do such as it can't store a state of some value. It can't do any operation related with memory. This is where sequential logic comes into play.
Sequential Logic might give a different output for the a given input at different times as it depends upon the previous values that have come.

2.1 Flip-Flops (FFs):

- Flip flops are one of the most simple memory elements (Stores only 1 bit).
- The output is the value stored inside the element, and this value only changes when the clock signal changes i.e when **clock edges(edge-triggered clocking)**.
- Clocks are essential in sequential logic to decide when an state of a element should be updated.
- In an edge-triggered methodology, either the rising edge or the falling edge of the clock causes a state change to occur. In this course we assume that only the positive edge triggering(rising edge) leads to change in value.

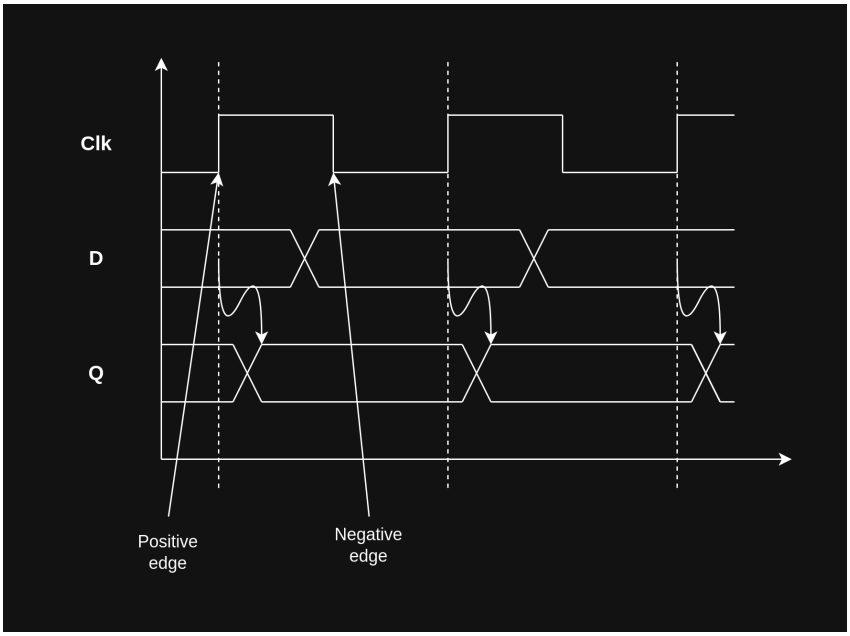


Figure 3: How data is changed in a positive edge-triggered flip flops

Flip Flops with write control: Flip flops can also be given an extra input control. **Write control** allows us to decide when we want the flip flop to change the value of it's state. The value of a state updates only when both the positive clock edge occurs and the write input is 1. (See Figure : 4)

2.2 Registers:

- A register is made up of multiple flip flops undergoing same operation. Each bit of data is stored in a flip-flop, so a register with 64 bits would consist of 64 flip-flops.
- The flip-flops in a register are usually synchronized with the same clock signal and all the flip flops of a register have the same write control. So when the state of register changes, all bits are updated together.

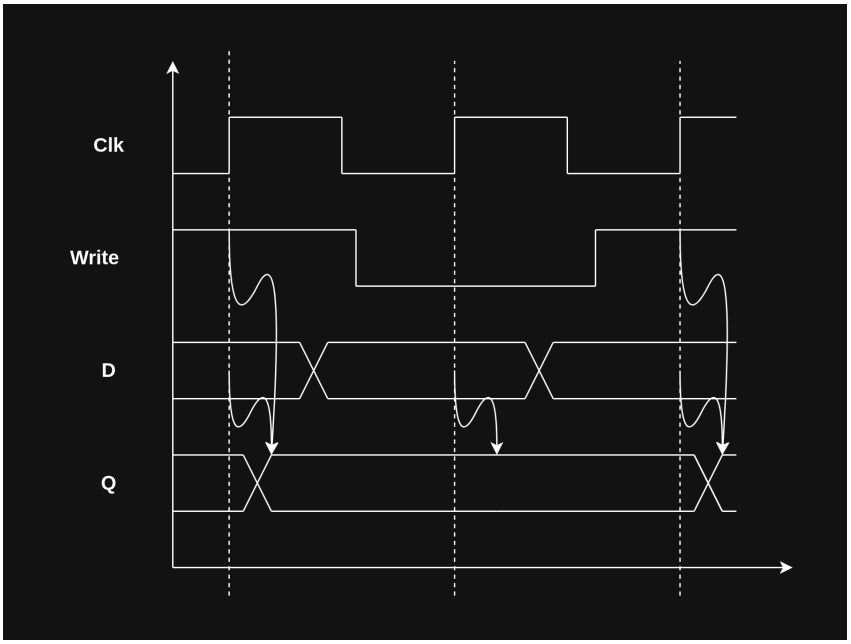


Figure 4: Write Control Signal for flip flops

2.3 Register File:

A register file consists of a set of registers that can be read and written. It is basically an array of registers. A register file typically consists of following interface:

- The **Addr** is an n-bit input and determined which register is to be accessed. For example, in RISC-V due to having 32 registers, Addr is a 5-bit input.
- **Write** is the write control to enable writing to the register selected by Addr
- **Clk** is the clock signal
- **D** and **Q** are input (write) and output (read) data respectively, of the same width as the width of each register.

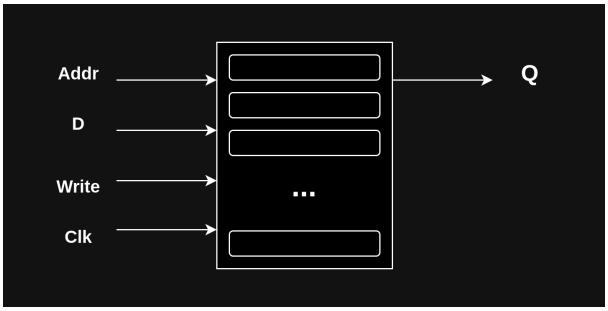


Figure 5: Register File

Internally,

- For reading a register, a multiplexor is used towards the output side to select the register value as per the Addr ⇒ The values of all registers are passed into a multiplexor and the control value decides which register to select.
- Similarly, on the input side, a decoder is used to select the appropriate register as per the Addr.

2.4 Memory:

Memory is a crucial component that allows the us to store and retrieve data during execution. Memory stores the intermediate data, instructions, and control information. Memory is generally RAM in our devices.

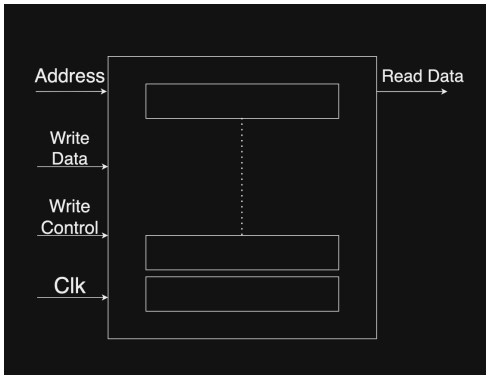


Figure 6: Memory

3 Clocking Methodology:

- Combinational logic transforms data during clock cycles and combinational logic have some delay.
- To ensure that the values written into the state elements on the active clock edge(positive edge) are valid, the clock must have a long enough period so that all the signals in the combinational logic block completes and stabilize.
- This constraint sets a lower bound on the length of the clock period(clock period is the time for one full cycle), which must be longer than delay caused by combinational logic block.

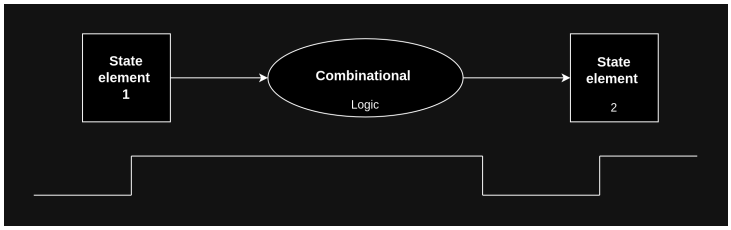


Figure 7: inputs come from a state element and outputs written into a state element

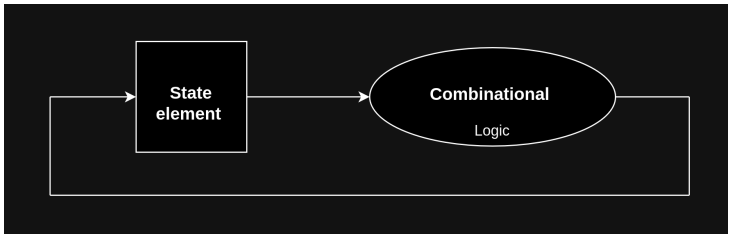


Figure 8: edge-triggered methodology allows a state element to read and written in the same clock cycle

- One advantage of an edge-triggered methodology is that it is valid to have a state element that can be used as both an input and output to the same combinational logic block because between the transitions, the value inside the flip-flop remains stable.

4 How the processor works?

4.1 How instructions are executed:

- Step 1: The address stored in PC is used to fetch the current instruction to be executed.
- Step 2: Then the registers and the immediate values are read from the instructions.
- Step 3: Then based on the given instruction, the ALU either calculates
  1. Arithmetic result ⇒ Then send the result to the registers
  2. Memory address for load ⇒ Send the address to memory and fetch the data from the address and then return it to the registers.
  3. Memory address for store ⇒ Send the address to memory and store the data to be stored at the calculated address.
  4. Branch comparison ⇒ If true then send the branch address to PC.
- Step 4: Increment the PC to PC+4 or the branch address.

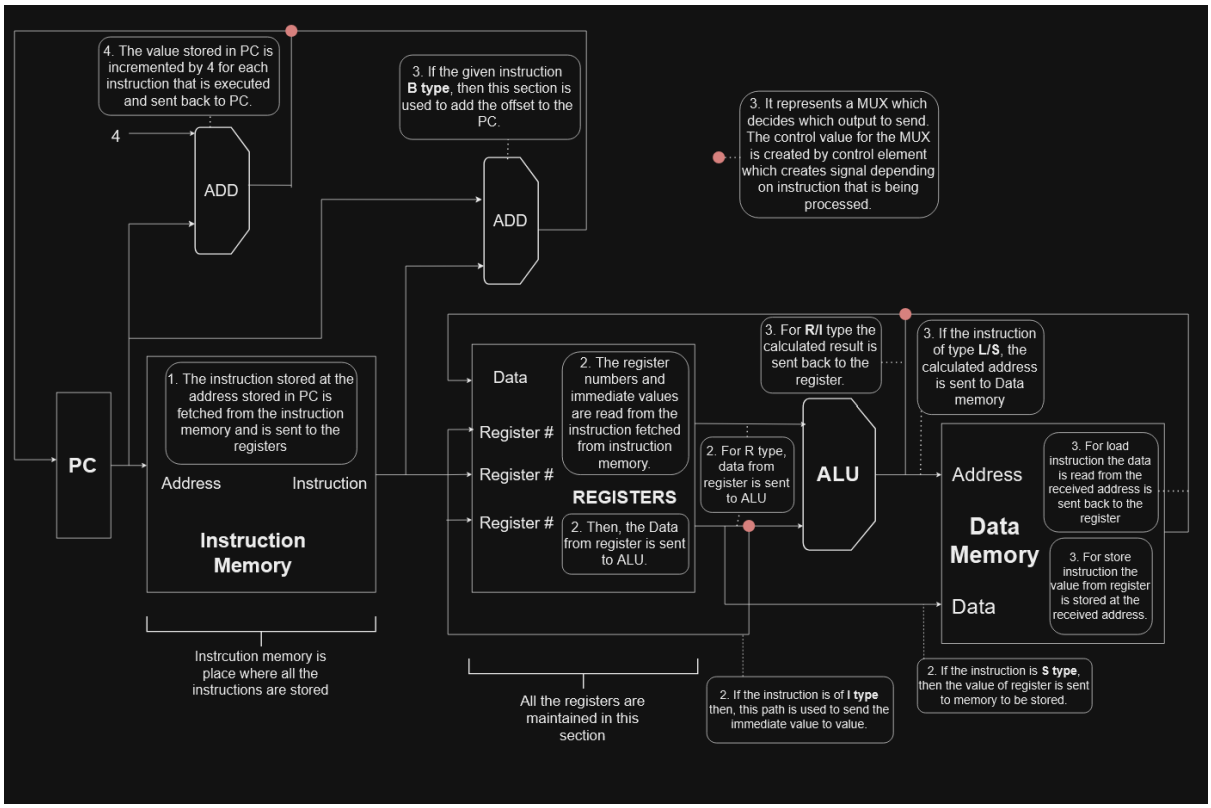


Figure 9: Basic Processor implementation

4.2 Datapath:

The elements that are used in processing data and address in CPU is:

- Register file
- ALU
- MUX

- Instruction Memory
- Data memory

4.3 Control

The control unit manages the overall operation of the processor by generating control signals that are used in MUXS throughout the circuit. It determines which operation the ALU should perform (add, subtract, etc.), what data to read/write to registers, when to read/write memory.